

# Vishal Dogra

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## Summary

Computer Science undergraduate at NIT Srinagar with hands-on full stack development experience — building production web platforms end-to-end with Next.js, React, Node.js, Express, and Python, backed by SQL/NoSQL data modeling across PostgreSQL and MongoDB. Security research internship at IIT Jammu building an automated SQL injection detection tool across multiple relational database engines. Strong foundation in DSA, OOP, and REST API design.

## Education

**National Institute of Technology, Srinagar** | *B.Tech – Computer Science & Engineering*

*Aug 2023 – Aug 2027*

Srinagar, J&K

**CGPA:** 7.94 / 10.0 **Coursework:** Data Structures & Algorithms, DBMS, Cloud Computing, Operating Systems, Computer Networks, OOP, Computer Organization & Architecture

## Experience

**Security Research Intern** | *IIT Jammu – Cyber Defense Lab*

*Dec 2025 – Feb 2026*

Supervisor: Dr. Gaurav Varshney

- Designed and implemented AEGIS Scanner v2.0, a modular Python-based SQL injection detection tool supporting 4 database engines (MySQL, PostgreSQL, MSSQL, Oracle) via a plugin-style, database-agnostic connector architecture.
- Implemented the design of SQL-driven detection logic (boolean-based, time-based, error-based, UNION-based queries) and automated payload/data generation pipelines, validated through manual testing across all supported backends.
- Directed implementation of session-based authentication (form login with CSRF protection) and structured vulnerability reporting for downstream consumption.
- Designed database-agnostic abstraction layers, enabling independent connector development and easy extension to new database targets.

**Web Developer (Contributor)** | *Rang-e-Chinar Cultural Fest, NIT Srinagar*

*September 2025 – October 2025*

Srinagar, J&K

- Independently implemented the participant ID image upload and verification flow using Cloudinary, integrated into a Next.js event registration platform serving 5,000+ participants.
- Handled client-to-cloud upload logic, storage configuration, and integration with the platform's existing data pipeline for participant verification.
- Collaborated within a cross-functional team under tight festival-launch deadlines.

## Projects

**Tomorrow's Datagram – College Activity Board** | *Next.js · TypeScript · PostgreSQL · SQL*

[github.com/vishalxdogra/activity-board2](https://github.com/vishalxdogra/activity-board2)

- Built secure roll number-based authentication (bcrypt, JWT, HTTP-only cookies) with verified-user and admin access controls, plus a Cloudinary-based student ID verification workflow with admin approval.
- Designed RESTful API routes and a relational PostgreSQL schema (Prisma) for activities, likes, comments, joins, and reports, with server-side Zod validation and Prisma transactions for reliable data handling.
- Implemented searchable, filterable activity feeds (genre/type filters, popularity sorting, capacity handling) and an admin moderation dashboard for approvals, rejections, and content reports.
- Stack:* Next.js 15, TypeScript, PostgreSQL, Prisma ORM, JWT, bcrypt, Cloudinary, Zod

**Wanderlust – Full Stack Accommodation Platform** | *Node.js · MongoDB · Express.js*

[github.com/vishalxdogra/wanderlust](https://github.com/vishalxdogra/wanderlust)

- Implemented secure authentication using Passport.js, deployed on Render with MongoDB Atlas, and designed RESTful API routes following MVC architecture principles.

**AEGIS Scanner v2.0 – SQL Injection Detection Tool** | *Python · SQL · PostgreSQL*

[github.com/vishalxdogra/Aegis-Scanner](https://github.com/vishalxdogra/Aegis-Scanner)

- Architected a modular Python-based SQL injection detection engine supporting automated vulnerability analysis across 4 database engines (MySQL, PostgreSQL, MSSQL, Oracle) with a plugin-style connector architecture.
- Built a multi-vector SQL query engine (boolean-based, time-based blind, error-based, UNION-based) with automated payload generation and session-based, CSRF-protected auth-aware access.
- Designed database-agnostic abstraction layers enabling independent connector development across all backends.
- Stack:* Python, SQL, PostgreSQL, MySQL, MSSQL, Oracle

## Technical Skills

**Data & SQL:** SQL, Data Modeling, Query Optimization, MongoDB

**Languages:** Python, C++, TypeScript, JavaScript

**Cloud & Platforms:** Docker, Render, Vercel

**Frameworks & Libraries:** Node.js, Express.js, Prisma ORM, React.js, Next.js

**Core CS:** Data Structures & Algorithms, DBMS, Operating Systems, OOP, REST APIs, CI/CD fundamentals

**Tools:** Git, GitHub, Postman